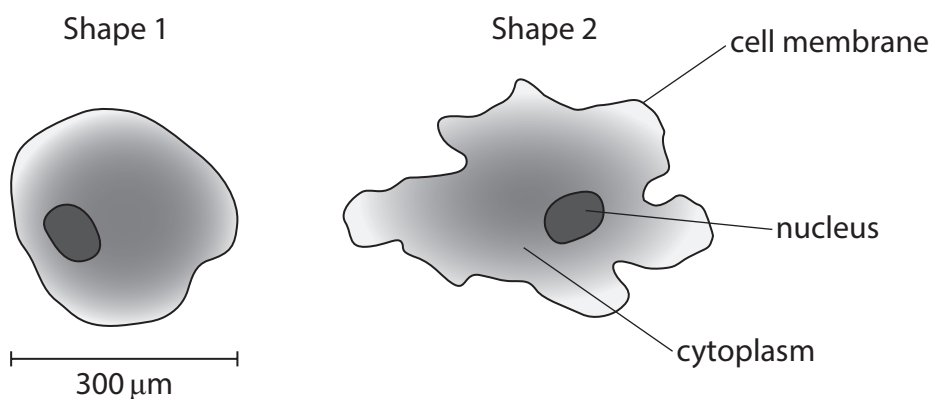


Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☐. If you change your mind about an answer, put a line through the box ☐ and then mark your new answer with a cross ☐.

- 1 Amoeba are single-celled organisms that live in water. They are able to change their shape.

The diagram shows two different shapes that one amoeba can make.



- (a) An amoeba changes shape by pushing cytoplasm against the cell membrane, forming long thin projections.

State the property of the cell membrane that enables the amoeba to change shape.

(1)

.....

.....

.....

- (b) Gas exchange in an amoeba takes place across its cell membrane.

- (i) The amoeba can be roughly spherical, as shown in Shape 1.

Which is the surface area of the amoeba in this shape?

Use the formula:

$$\text{surface area of a sphere} = 4\pi r^2$$

(1)

- ☐ A $90\,000\pi\,\mu\text{m}^2$
- ☐ B $90\,000\pi\,\mu\text{m}^3$
- ☐ C $360\,000\pi\,\mu\text{m}^2$
- ☐ D $360\,000\pi\,\mu\text{m}^3$



